# Exit Ticket · reflect and retrieve

## Exit Ticket

A basic Ethernet switch chooses internet routes using destination website names.

## Use key vocabulary from memory

1. Explain switch in your own words.

2. Explain router in your own words.

3. Explain access point in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Lesson 148 · Network Hardware: Switches, Routers and Access Points

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What does a firewall generally use to permit or block a connection?

### Q2

What data structure follows FIFO?

### Q3

What is the result of an AND gate with inputs 1 and 0?

### Q4

What is a foreign key used for?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Local connections — A switch forwards Ethernet frames using learned MAC addresses.

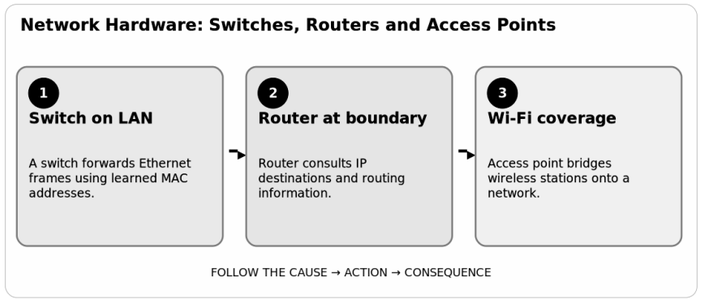
## Core idea 2

Across networks — Router consults IP destinations and routing information.

## Core idea 3

Wireless access — Access point bridges wireless stations onto a network.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Design a network for two labs and a library with wireless devices and an internet connection. Label device placement and traffic path. [8]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Differentiate switch, router and access point with their forwarding roles. [6]

## Question 2

For a PC accessing a website, explain why both MAC and IP addresses are useful. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

switch — Connects local devices and forwards data-link frames.

router — Forwards network-layer IP packets between networks.

access point — Connects wireless devices to a LAN.

NIC — Network interface controller for network connectivity.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Two school laptops on one wired LAN share a file; one then accesses a website on another network. Trace the hardware.

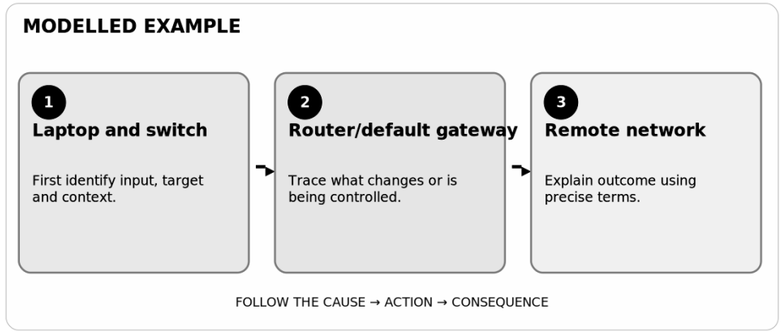
## Follow the worked method

1. Laptop and switch — Identify the facts and make an initial prediction.

2. Router/default gateway — Explain the mechanism or trace the intermediate stage.

3. Remote network — Within the LAN, NICs form frames and the switch forwards using MAC addresses. To reach another IP network, traffic is sent to a router/default gateway that forwards packets. The gateway rewrites the link framing per next hop.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Wired LAN works
Identify the facts and make an initial prediction.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.